Experiment No. 9

Experiment Name - Data Storytelling and Business Insight Generation using Interactive Visualizations

Steps

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COURSE : B.Tech CSE

SECTION - B

SUBJECT : Mastery in Data Analytics and Visualizations and Career Advancement

ROLL NO. : 14

CU ID : CU24250231

1. Select a real-world dataset such as the Global Superstore Sales Dataset, Netflix Dataset, World Happiness Report, or COVID19 Dataset.

```
# Selecting a Real-World Dataset
from google.colab import files
uploaded = files.upload()

import os
print(os.listdir())

dataset_name = "Global Superstore Sales Dataset"

print("Selected Dataset:", dataset_name)
print("Dataset selected successfully!")
```

```
Choose Files Global Superstore.csv
Global Superstore.csv (text/csv) - 12444106 bytes, last modified: 10/5/2026 - 100% done
Saving Global Superstore.csv to Global Superstore.csv
['.config', 'Global Superstore.csv', 'sample_data' ]
Selected Dataset: Global Superstore Sales Dataset
Dataset selected successfully!
```

2. Analyze the dataset to identify the primary business problem or analytical objective.

Business Problem:

The primary business problem is to analyze the Global Superstore Sales Dataset to understand sales and profit performance, identify high-performing product categories and regions, and detect loss-making products or sub-categories.

Analytical Objectives:

1. To analyze the overall sales and profit of the business.
2. To identify the best-performing product categories based on sales and profit.
3. To compare sales and profitability across different regions.
4. To identify loss-making products and sub-categories.
5. To develop data-driven recommendations to improve business profitability and support strategic decision-making.

3. Perform exploratory data analysis to discover significant trends, patterns, correlations, and anomalies.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px

sns.set_theme(style="whitegrid")

# Load the uploaded dataset
df = pd.read_csv("Global Superstore.csv", encoding="latin1")

print("Dataset loaded successfully!")
print("Dataset Shape:", df.shape)

print("\nFirst 5 Rows:")
display(df.head())

print("\nColumn Names:")
print(df.columns.tolist())
```

```
Dataset loaded successfully!
Dataset Shape: (51290, 27)

First 5 Rows:
```

	Category	City	Country	Customer ID	Customer Name	Discount	Market	ji_lu-shu	Order Date	Order ID	...	Sales	Segment	Ship Date	Ship Mode	Shipping Cost	Sta
0	Office Supplies	Los Angeles	United States	LS-172304	Lycoris Saunders	0.0	US	1	00:00.0	CA-2011-130813	...	19	Consumer	00:00.0	Second Class	4.37	Californ
1	Office Supplies	Los Angeles	United States	MV-174854	Mark Van Huff	0.0	US	1	00:00.0	CA-2011-148614	...	19	Consumer	00:00.0	Standard Class	0.94	Californ
2	Office Supplies	Los Angeles	United States	CS-121304	Chad Sievert	0.0	US	1	00:00.0	CA-2011-118962	...	21	Consumer	00:00.0	Standard Class	1.81	Californ
3	Office Supplies	Los Angeles	United States	CS-121304	Chad Sievert	0.0	US	1	00:00.0	CA-2011-118962	...	111	Consumer	00:00.0	Standard Class	4.59	Californ
4	Office Supplies	Los Angeles	United States	AP-109154	Arthur Prichep	0.0	US	1	00:00.0	CA-2011-146969	...	6	Consumer	00:00.0	Standard Class	1.32	Californ

5 rows × 27 columns

```
Column Names:
['Category', 'City', 'Country', 'Customer ID', 'Customer Name', 'Discount', 'Market', 'ji_lu-shu', 'Order Date', 'Order ID', 'Order Priority', 'Pr
```

```
# Check dataset information
print("Dataset Information:")
df.info()

print("\nMissing Values in Each Column:")
print(df.isnull().sum())

print("\nTotal Missing Values:", df.isnull().sum().sum())

print("\nTotal Duplicate Rows:", df.duplicated().sum())
```

```
 5   Discount        51290 non-null  float64
 6   Market          51290 non-null  object
 7   ji_lu-shu       51290 non-null  int64
 8   Order Date      51290 non-null  object
 9   Order ID        51290 non-null  object
 10  Order Priority  51290 non-null  object
 11  Product ID      51290 non-null  object
 12  Product Name    51290 non-null  object
 13  Profit          51290 non-null  float64
 14  Quantity        51290 non-null  int64
 15  Region          51290 non-null  object
 16  Row ID          51290 non-null  int64
 17  Sales           51290 non-null  int64
 18  Segment         51290 non-null  object
 19  Ship Date       51290 non-null  object
 20  Ship Mode       51290 non-null  object
 21  Shipping Cost   51290 non-null  float64
 22  State           51290 non-null  object
 23  Sub-Category    51290 non-null  object
 24  Year            51290 non-null  int64
 25  Market2         51290 non-null  object
 26  weeknum         51290 non-null  int64
dtypes: float64(3), int64(6), object(18)
memory usage: 10.6+ MB

Missing Values in Each Column:
Category          0
City              0
Country           0
Customer ID       0
Customer Name     0
Discount          0
Market            0
ji_lu-shu         0
Order Date        0
Order ID          0
Order Priority    0
Product ID        0
Product Name      0
Profit            0
Quantity          0
Region            0
Row ID            0
Sales             0
```

```
Total Missing Values: 0

Total Duplicate Rows: 0
```

```
# Create a working copy of the dataset
df_clean = df.copy()

# Remove exact duplicate rows
df_clean = df_clean.drop_duplicates().copy()

# Remove rows missing essential analysis values
df_clean = df_clean.dropna(
    subset=["Sales", "Profit", "Quantity", "Category", "Region"]
).copy()

# Convert numerical columns to numeric format
numeric_columns = [
    "Sales", "Profit", "Quantity", "Discount", "Shipping Cost"
]

for col in numeric_columns:
    df_clean[col] = pd.to_numeric(df_clean[col], errors="coerce")

# Remove rows where essential numeric values are invalid
df_clean = df_clean.dropna(
    subset=["Sales", "Profit", "Quantity"]
).copy()

print("Original Dataset Shape:", df.shape)
print("Cleaned Dataset Shape:", df_clean.shape)
print("Missing Values Remaining in Essential Columns:")
print(df_clean[["Sales", "Profit", "Quantity"]].isnull().sum())
```

```
Original Dataset Shape: (51290, 27)
Cleaned Dataset Shape: (51290, 27)
Missing Values Remaining in Essential Columns:
Sales       0
Profit      0
Quantity    0
dtype: int64
```

```
# Generate descriptive statistics
print("Statistical Summary:")
display(df_clean.describe())

print("\nUnique Product Categories:")
print(df_clean["Category"].nunique())

print("\nUnique Regions:")
print(df_clean["Region"].nunique())

print("\nUnique Countries:")
print(df_clean["Country"].nunique())

print("\nUnique Customers:")
print(df_clean["Customer ID"].nunique())

print("\nUnique Orders:")
print(df_clean["Order ID"].nunique())
```

Statistical Summary:

	Discount	ji_lu-shu	Profit	Quantity	Row ID	Sales	Shipping Cost	Year	weeknum
count	51290.000000	51290.0	51290.000000	51290.000000	51290.00000	51290.000000	51290.000000	51290.000000	51290.000000
mean	0.142908	1.0	28.610982	3.476545	25645.50000	246.498440	26.375818	2012.777208	31.287112
std	0.212280	0.0	174.340972	2.278766	14806.29199	487.567175	57.296810	1.098931	14.429795
min	0.000000	1.0	-6599.978000	1.000000	1.00000	0.000000	0.002000	2011.000000	1.000000
25%	0.000000	1.0	0.000000	2.000000	12823.25000	31.000000	2.610000	2012.000000	20.000000
50%	0.000000	1.0	9.240000	3.000000	25645.50000	85.000000	7.790000	2013.000000	33.000000
75%	0.200000	1.0	36.810000	5.000000	38467.75000	251.000000	24.450000	2014.000000	44.000000
max	0.850000	1.0	8399.976000	14.000000	51290.00000	22638.000000	933.570000	2014.000000	53.000000

```
Unique Product Categories:
3

Unique Regions:
13

Unique Countries:
147

Unique Customers:
4873

Unique Orders:
25035
```

```
# Overall sales and profit analysis
total_sales = df_clean["Sales"].sum()
```

```
total_profit = df_clean["Profit"].sum()
total_quantity = df_clean["Quantity"].sum()
total_orders = df_clean["Order ID"].nunique()
total_customers = df_clean["Customer ID"].nunique()

profit_margin = (
    (total_profit / total_sales) * 100
    if total_sales != 0 else 0
)

print("========== BUSINESS OVERVIEW ==========")
print(f"Total Sales: {total_sales:,.2f}")
print(f"Total Profit: {total_profit:,.2f}")
print(f"Total Quantity Sold: {total_quantity:,}")
print(f"Total Unique Orders: {total_orders:,}")
print(f"Total Unique Customers: {total_customers:,}")
print(f"Overall Profit Margin: {profit_margin:.2f}%")
```

```
========== BUSINESS OVERVIEW ==========
Total Sales: 12,642,905.00
Total Profit: 1,467,457.29
Total Quantity Sold: 178,312
Total Unique Orders: 25,035
Total Unique Customers: 4,873
Overall Profit Margin: 11.61%
```

```
# Category-wise analysis
category_analysis = (
    df_clean.groupby("Category", as_index=False)
    .agg(
        Total_Sales=("Sales", "sum"),
        Total_Profit=("Profit", "sum"),
        Total_Quantity=("Quantity", "sum")
    )
    .sort_values("Total_Sales", ascending=False)
)

print("Category-wise Business Performance:")
display(category_analysis)

# Region-wise analysis
region_analysis = (
    df_clean.groupby("Region", as_index=False)
    .agg(
        Total_Sales=("Sales", "sum"),
        Total_Profit=("Profit", "sum"),
        Total_Quantity=("Quantity", "sum")
    )
    .sort_values("Total_Profit", ascending=False)
)

print("\nRegion-wise Business Performance:")
display(region_analysis)
```

```
Category-wise Business Performance:
```

	Category	Total_Sales	Total_Profit	Total_Quantity
2	Technology	4744691	663778.73318	35176
0	Furniture	4110884	285204.72380	34954
1	Office Supplies	3787330	518473.83430	108182

```
Region-wise Business Performance:
```

	Region	Total_Sales	Total_Profit	Total_Quantity
3	Central	2822399	311403.98164	41762
7	North	1248192	194597.95252	18021
8	North Asia	848349	165578.42100	8851
10	South	1600960	140355.76618	25206
4	Central Asia	752839	132480.18700	7715
9	Oceania	1100207	120089.11200	12838
12	West	725514	108418.44890	12266
6	East	678834	91522.78000	10618
0	Africa	783776	88871.63100	10564
5	EMEA	806184	43897.97100	11517
2	Caribbean	324281	34571.32104	6299
11	Southeast Asia	884438	17852.32900	11822
1	Canada	66932	17817.39000	833

Next steps: Generate code with `category_analysis` | New interactive sheet | Generate code with `region_analysis` | New interactive sheet

```
# Sub-category analysis
subcategory_analysis = (
```

```
    df_clean.groupby("Sub-Category", as_index=False)
    .agg(
        Total_Sales=("Sales", "sum"),
        Total_Profit=("Profit", "sum"),
        Total_Quantity=("Quantity", "sum")
    )
    .sort_values("Total_Profit", ascending=True)
)

print("Five Lowest-Profit Sub-Categories:" )
display(subcategory_analysis.head(5))

print("\nFive Highest-Profit Sub-Categories:" )
display(subcategory_analysis.tail(5).sort_values(
    "Total_Profit", ascending=False
))
```

```
Five Lowest-Profit Sub-Categories:
```

	Sub-Category	Total_Sales	Total_Profit	Total_Quantity
16	Tables	757034	-64083.3887	3083
8	Fasteners	83254	11525.4241	8390
10	Labels	73433	15010.5120	9322
15	Supplies	243090	22583.2631	8543
7	Envelopes	170926	29601.1163	8380

```
Five Highest-Profit Sub-Categories:
```

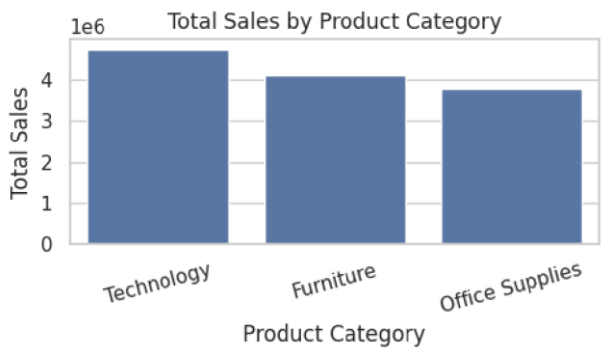
	Sub-Category	Total_Sales	Total_Profit	Total_Quantity
6	Copiers	1509439	258567.54818	7454
13	Phones	1706874	216717.00580	11870
4	Bookcases	1466559	161924.41950	8310
1	Appliances	1011081	141680.58940	6078
5	Chairs	1501682	140396.26750	12336

4. Develop a set of meaningful visualizations such as bar charts, line graphs, heatmaps, and dashboards to support the analysis.

```
# Visualize sales by product category
plt.figure(figsize=(5, 3))

sns.barplot(
    data=category_analysis ,
    x="Category",
    y="Total_Sales"
)

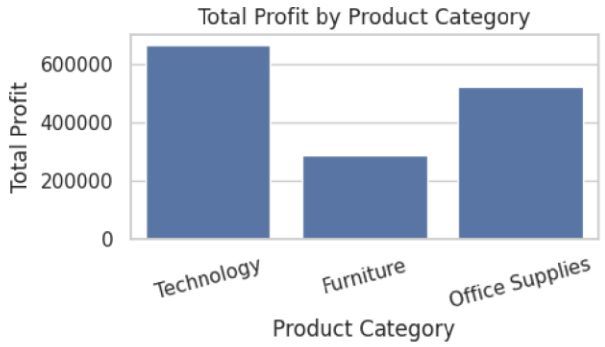
plt.title("Total Sales by Product Category" )
plt.xlabel("Product Category" )
plt.ylabel("Total Sales")
plt.xticks(rotation=15)
plt.tight_layout()
plt.show()
```




```
# Visualize profit by product category
plt.figure(figsize=(5, 3))

sns.barplot(
    data=category_analysis ,
    x="Category",
    y="Total_Profit"
)

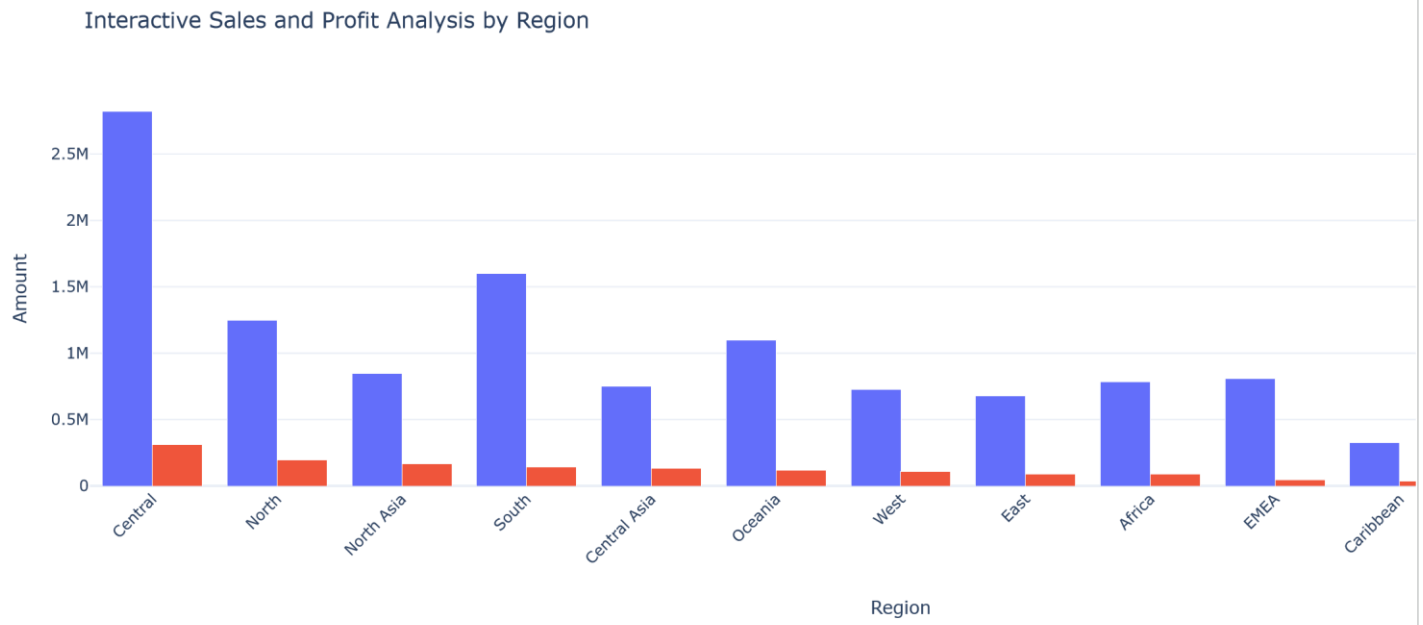
plt.title("Total Profit by Product Category" )
plt.xlabel("Product Category" )
plt.ylabel("Total Profit")
plt.xticks(rotation=15)
plt.tight_layout()
plt.show()
```




```
# Interactive regional performance visualization
fig = px.bar(
    region_analysis,
    x="Region",
    y=["Total_Sales", "Total_Profit"],
    barmode="group",
    title="Interactive Sales and Profit Analysis by Region",
    labels={
        "value": "Amount",
        "variable": "Business Metric",
        "Region": "Region"
    },
    hover_data=["Total_Quantity"]
)

fig.update_layout(
    xaxis_tickangle=-45,
    template="plotly_white"
)

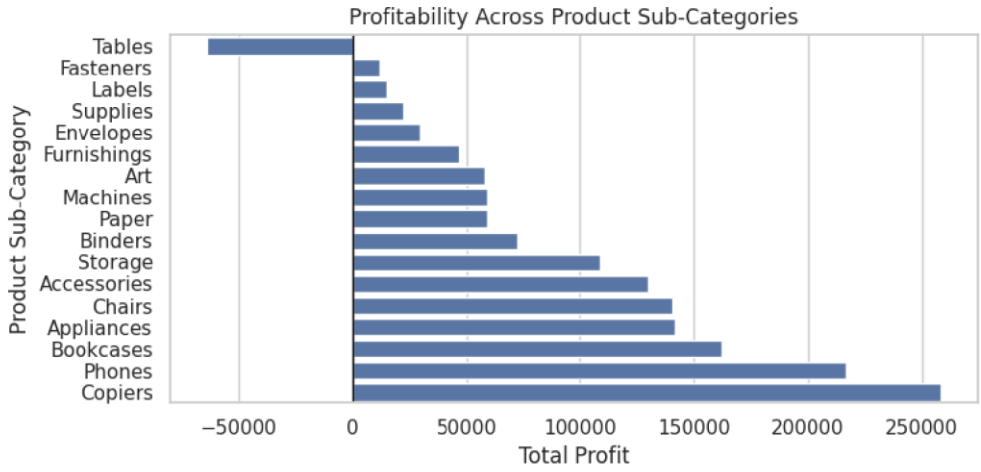
fig.show()
```




```
# Identify loss-making and profitable sub-categories
plt.figure(figsize=(8, 4))

sns.barplot(
    data=subcategory_analysis,
    x="Total_Profit",
    y="Sub-Category"
)

plt.axvline(0, color="black", linewidth=1)
plt.title("Profitability Across Product Sub-Categories")
plt.xlabel("Total Profit")
plt.ylabel("Product Sub-Category")
plt.tight_layout()
plt.show()
```




```
# Analyze annual sales trends
year_analysis = (
    df_clean.groupby("Year", as_index=False)
    .agg(
        Total_Sales=("Sales", "sum"),
        Total_Profit=("Profit", "sum")
    )
    .sort_values("Year")
)

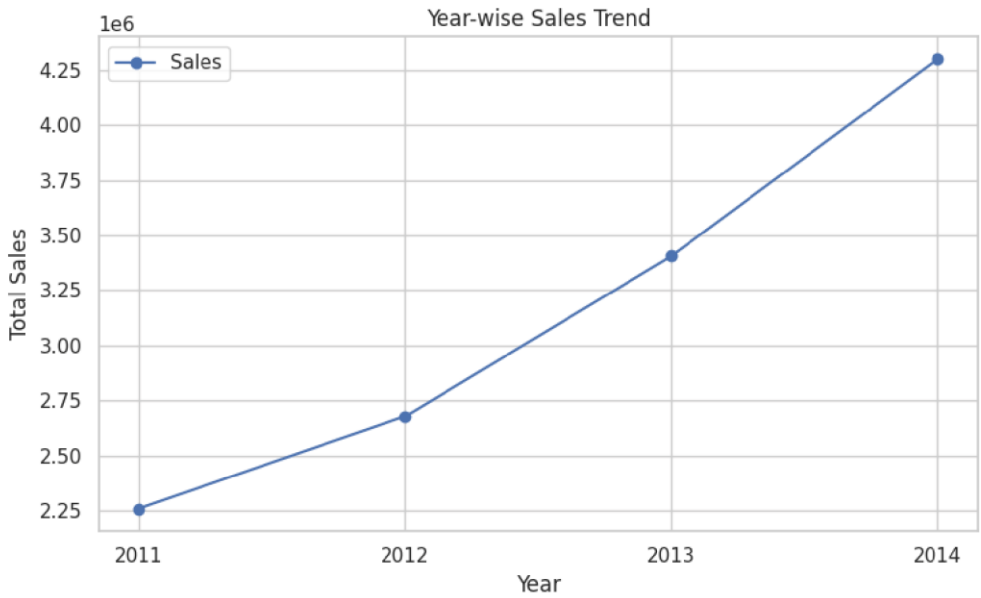
display(year_analysis)

plt.figure(figsize=(8, 5))

plt.plot(
    year_analysis["Year"],
    year_analysis["Total_Sales"],
    marker="o",
    label="Sales"
)

plt.title("Year-wise Sales Trend")
plt.xlabel("Year")
plt.ylabel("Total Sales")
plt.xticks(year_analysis["Year"])
plt.legend()
plt.tight_layout()
plt.show()
```

	Year	Total_Sales	Total_Profit
0	2011	2259511	248940.81154
1	2012	2677493	307415.27910
2	2013	3405860	406935.23018
3	2014	4300041	504165.97046



Next steps: Generate code with year_analysis | New interactive sheet

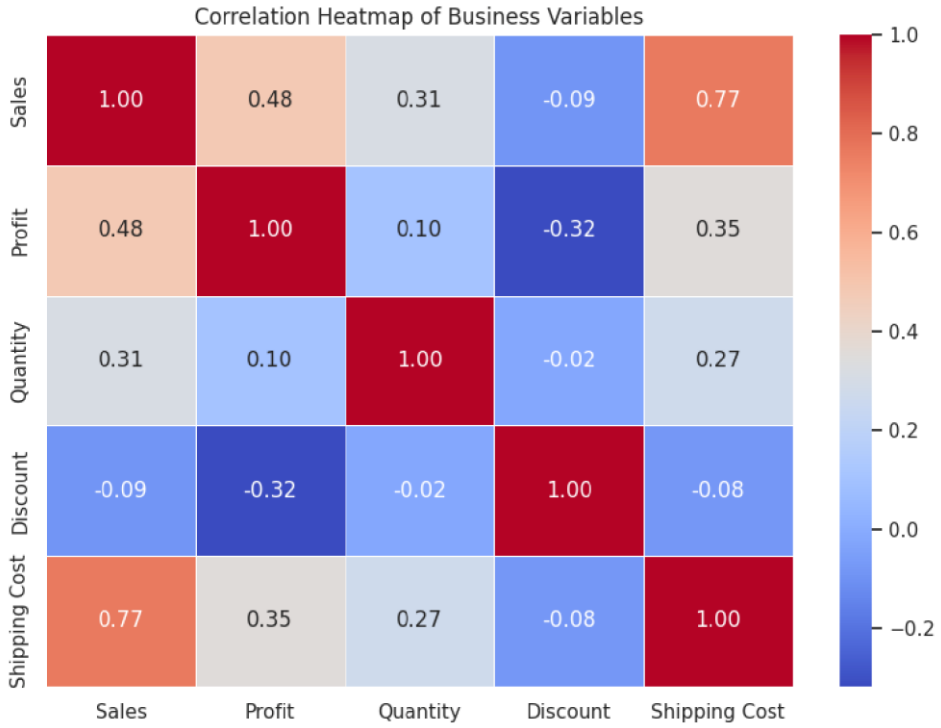
```
# Correlation heatmap
correlation_columns = [
    "Sales", "Profit", "Quantity", "Discount", "Shipping Cost"
]
```

```
correlation_matrix = df_clean [
    correlation_columns
].corr()

plt.figure(figsize=(8, 6))

sns.heatmap(
    correlation_matrix ,
    annot=True,
    cmap="coolwarm",
    fmt=".2f",
    linewidths=0.5
)

plt.title("Correlation Heatmap of Business Variables" )
plt.tight_layout()
plt.show()
```



5. Identify Key Performance Indicators (KPIs) and highlight important business metrics relevant to the selected dataset.

6. Arrange the visualizations in a logical sequence to create a coherent data story that explains the problem, analysis, findings,and recommendations.

```
# Key Performance Indicators (KPIs)
total_sales = df_clean ["Sales"].sum()
total_profit = df_clean ["Profit"].sum()

total_orders = df_clean ["Order ID"].nunique()
total_customers = df_clean ["Customer ID"].nunique()

total_quantity = df_clean ["Quantity"].sum()

profit_margin = (
    total_profit / total_sales *  100
    if total_sales != 0 else 0
)

average_order_value =  (
    total_sales / total_orders
    if total_orders != 0 else 0
)

loss_making_subcategories =  (
    subcategory_analysis [
        subcategory_analysis ["Total_Profit"] < 0
    ]["Sub-Category"].tolist()
)

print("KEY PERFORMANCE INDICATORS:" )
print(f"Total Sales             :  {total_sales:,.2f}")
print(f"Total Profit            :  {total_profit:,.2f}")
print(f"Profit Margin           :  {profit_margin:.2f}%")
print(f"Total Orders            :  {total_orders:,}")
print(f"Total Customers         :  {total_customers:,}")
print(f"Total Quantity Sold     :  {total_quantity:,}")
print(f"Average Order Value     :  {average_order_value:,.2f}")
print(f"Loss-making Sub-Categories:  {loss_making_subcategories }")
```

```
KEY PERFORMANCE INDICATORS:
Total Sales             : 12,642,905.00
Total Profit            : 1,467,457.29
Profit Margin           : 11.61%
```

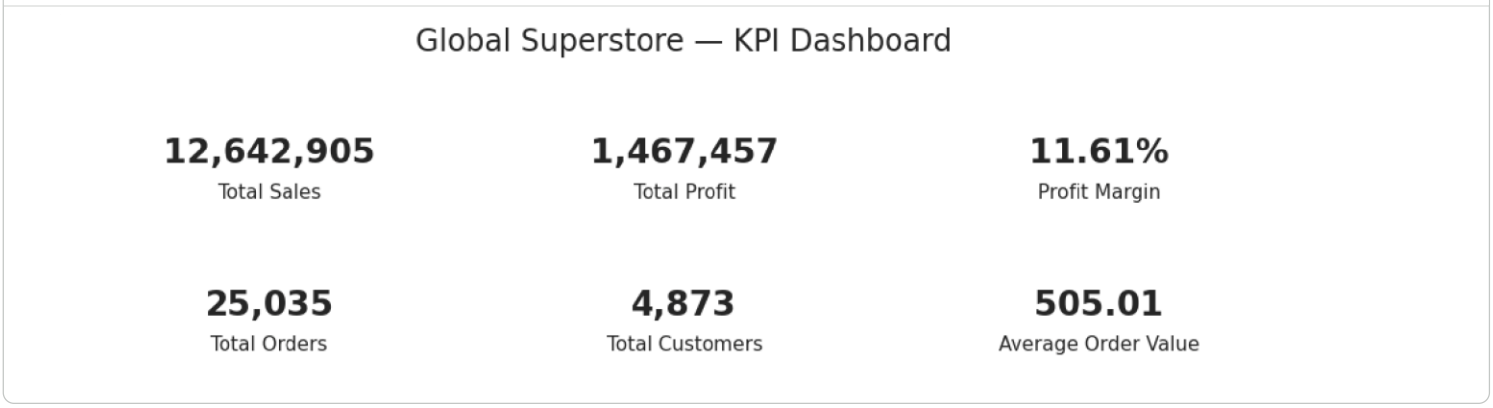
```
Total Orders            : 25,035
Total Customers         : 4,873
Total Quantity Sold     : 178,312
Average Order Value     : 505.01
Loss-making Sub-Categories: ['Tables']
```

```
# Visual KPI Dashboard
kpis = {
    "Total Sales": f"{total_sales:,.0f}",
    "Total Profit": f"{total_profit:,.0f}",
    "Profit Margin": f"{profit_margin:.2f}%",
    "Total Orders": f"{total_orders:,}",
    "Total Customers": f"{total_customers:,}",
    "Average Order Value": f"{average_order_value:,.2f}"
}

fig, axes = plt.subplots(2, 3, figsize=(10, 3))
axes = axes.flatten()

for ax, (label, value) in zip(axes, kpis.items()):
    ax.text(
        0.5, 0.55, value,
        ha="center", va="center",
        fontsize=19, fontweight="bold"
    )
    ax.text(
        0.5, 0.25, label,
        ha="center", va="center",
        fontsize=11
    )
    ax.set_axis_off()

plt.suptitle("Global Superstore – KPI Dashboard" , fontsize=17)
plt.tight_layout()
plt.show()
```



Data Story: Global Superstore Sales Analysis

The visualizations are arranged in a logical sequence to communicate the business problem, analytical findings, and recommendations effectively.

1. **Business Overview:** KPI cards showing total sales, total profit, profit margin, and total orders provide an overview of business performance.
2. **Product Performance:** Bar charts comparing sales and profit across product categories help identify the best-performing categories.
3. **Regional Performance:** Regional sales and profit charts highlight differences in performance across geographical regions.
4. **Profitability Challenges:** A sub-category profitability chart identifies products or sub-categories that generate losses.
5. **Sales Trends:** A line chart showing year-wise sales helps identify changes in business performance over time.
6. **Business Recommendations:** Based on the findings, the business can review pricing strategies, discounts, operating costs, and product profitability to improve overall performance.

Conclusion:

This sequence transforms individual visualizations into a coherent data story. It begins with an overview of business performance, investigates product and regional results, identifies challenges, and concludes with actionable recommendations for better decisionmaking.

7. **Interpret the visualizations and identify major insights, opportunities, risks, or performance gaps.**

```
# Generate Data-Driven Insights
best_sales_category = category_analysis.loc[
    category_analysis["Total_Sales"].idxmax()
]

best_profit_category = category_analysis.loc[
    category_analysis["Total_Profit"].idxmax()
```

```
]

best_region = region_analysis.loc[
    region_analysis["Total_Profit"].idxmax()
]

lowest_profit_region = region_analysis.loc[
    region_analysis["Total_Profit"].idxmin()
]

lowest_profit_subcategory = subcategory_analysis.iloc[0]

print("KEY BUSINESS INSIGHTS:")

print(
    f"\n1. Highest Sales Category: {best_sales_category['Category']}"
)
print(
    f"   Sales: {best_sales_category['Total_Sales']:,.2f}"
)

print(
    f"\n2. Highest Profit Category: {best_profit_category['Category']}"
)
print(
    f"   Profit: {best_profit_category['Total_Profit']:,.2f}"
)

print(f"\n3. Most Profitable Region: {best_region['Region']}")
print(f"   Profit: {best_region['Total_Profit']:,.2f}")

print(f"\n4. Lowest-Profit Region: {lowest_profit_region['Region']}")
print(f"   Profit: {lowest_profit_region['Total_Profit']:,.2f}")

print(
    f"\n5. Lowest-Profit Sub-Category: "
    f"{lowest_profit_subcategory['Sub-Category']}"
)
print(
    f"   Profit: {lowest_profit_subcategory['Total_Profit']:,.2f}"
)

print(f"\n6. Overall Profit Margin: {profit_margin:.2f}%")

if year_analysis["Year"].nunique() >= 2:
    first_year = year_analysis.iloc[0]
    last_year = year_analysis.iloc[-1]

    change = last_year["Total_Sales"] - first_year["Total_Sales"]

    print(
        f"\n7. Sales Change from {first_year['Year']} "
        f"to {last_year['Year']}: {change:,.2f}"
    )

    if first_year["Total_Sales"] != 0:
        percentage_change = (
            change / first_year["Total_Sales"] * 100
        )
        print(f"   Percentage Change: {percentage_change:.2f}%")
```

```
KEY BUSINESS INSIGHTS:

1. Highest Sales Category: Technology
   Sales: 4,744,691.00

2. Highest Profit Category: Technology
   Profit: 663,778.73

3. Most Profitable Region: Central
   Profit: 311,403.98

4. Lowest-Profit Region: Canada
   Profit: 17,817.39

5. Lowest-Profit Sub-Category: Tables
   Profit: -64,083.39

6. Overall Profit Margin: 11.61%

7. Sales Change from 2011.0 to 2014.0: 2,040,530.00
   Percentage Change: 90.31%
```

```
# Investigate the relationship between discounts and profit
discount_analysis = (
    df_clean.groupby("Discount", as_index=False)
    .agg(
        Total_Sales=("Sales", "sum"),
        Total_Profit=("Profit", "sum"),
        Number_of_Records=("Profit", "size")
    )
    .sort_values("Discount")
)
```

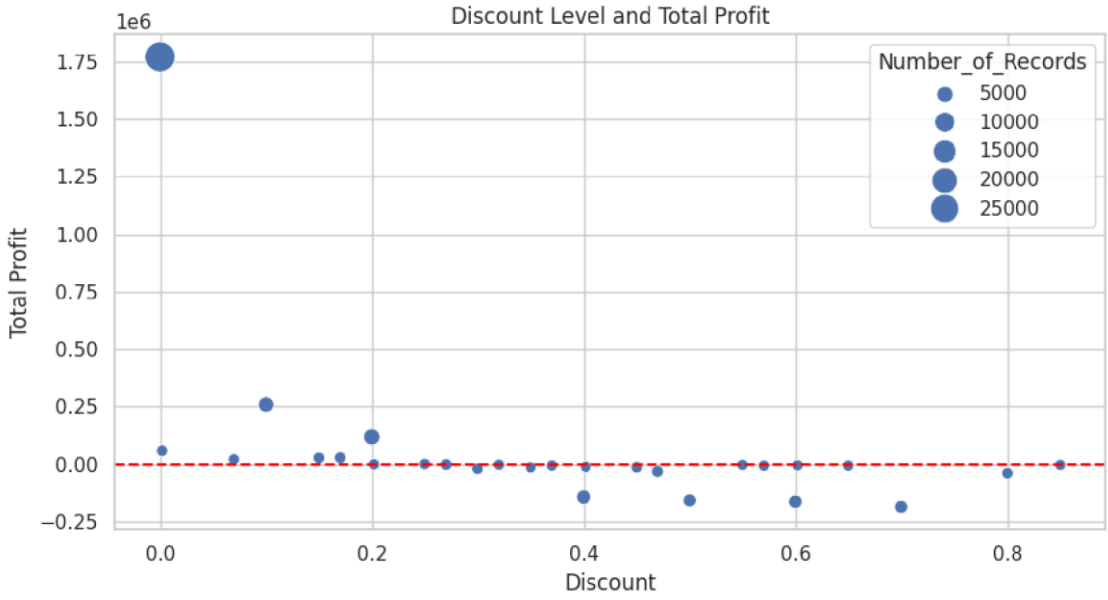
```
display(discount_analysis)

plt.figure(figsize=(9, 5))

sns.scatterplot(
    data=discount_analysis,
    x="Discount",
    y="Total_Profit",
    size="Number_of_Records",
    sizes=(40, 300)
)

plt.axhline(0, color="red", linestyle="--")
plt.title("Discount Level and Total Profit")
plt.xlabel("Discount")
plt.ylabel("Total Profit")
plt.tight_layout()
plt.show()
```

	Discount	Total_Sales	Total_Profit	Number_of_Records
0	0.000	6992734	1.770695e+06	29009
1	0.002	261395	5.797658e+04	461
2	0.070	121620	2.114850e+04	150
3	0.100	1579618	2.590642e+05	4068
4	0.150	306037	2.737590e+04	541
5	0.170	247404	2.816307e+04	735
6	0.200	1203855	1.177159e+05	4998
7	0.202	16213	-5.952727e+02	41
8	0.250	88063	8.005875e+02	198
9	0.270	89351	-1.675079e+03	388
10	0.300	188913	-1.968585e+04	340
11	0.320	14491	-2.391138e+03	27
12	0.350	95704	-1.416965e+04	122
13	0.370	31615	-5.806202e+03	74
14	0.400	559557	-1.437485e+05	3177
15	0.402	33737	-1.143045e+04	104
16	0.450	33397	-1.360679e+04	327
17	0.470	83432	-3.116225e+04	725
18	0.500	324145	-1.586299e+05	1633
19	0.550	6308	-3.150672e+03	10
20	0.570	8144	-6.313548e+03	12
21	0.600	184586	-1.639547e+05	2006
22	0.602	4977	-4.905419e+03	23
23	0.650	6580	-6.221966e+03	17
24	0.700	139770	-1.863505e+05	1786
25	0.800	20462	-3.861623e+04	316
26	0.850	797	-3.068658e+03	2



Next steps: Generate code with `discount_analysis` New interactive sheet

8. **Prepare actionable recommendations based on the analytical findings to support business decision-making.**

ACTIONABLE BUSINESS RECOMMENDATIONS:

1. **Product Strategy:** Focus on high-performing product categories to improve sales and profitability.
2. **Regional Strategy:** Improve the performance of low-profit regions by reviewing pricing, costs, and customer demand.
3. **Loss Prevention:** Review loss-making products and sub-categories to identify the reasons for losses.
4. **Discount Management:** Optimize discounts to increase sales without reducing profitability.
5. **Performance Monitoring:** Regularly monitor sales, profit, profit margin, and other KPIs to support better business decisions.

9. **Present the complete data story using presentation slides, Tableau, Power BI, or a Jupyter Notebook.**

```
# Interactive Business Dashboard
import plotly.graph_objects  as go
from plotly.subplots  import make_subplots

fig = make_subplots (
    rows=2,
    cols=2,
    subplot_titles= (
        "Sales by Category" ,
        "Profit by Category" ,
        "Sales by Region" ,
        "Profit by Sub-Category"
    ),
    specs=[
        [{"type": "bar"}, {"type": "bar"}],
        [{"type": "bar"}, {"type": "bar"}]
    ]
)

# Sales by category
fig.add_trace (
    go.Bar (
        x=category_analysis ["Category"],
        y=category_analysis ["Total_Sales"],
        name= "Sales"
    ),
    row=1, col=1
)

# Profit by category
fig.add_trace (
    go.Bar (
        x=category_analysis ["Category"],
        y=category_analysis ["Total_Profit"],
        name= "Profit"
    ),
    row=1, col=2
)

# Sales by region
fig.add_trace (
    go.Bar (
        x=region_analysis ["Region"],
        y=region_analysis ["Total_Sales"],
        name= "Regional Sales"
    ),
    row=2, col=1
)

# Profit by sub-category
fig.add_trace (
    go.Bar (
        x=subcategory_analysis ["Sub-Category"],
        y=subcategory_analysis ["Total_Profit"],
        name= "Sub-Category Profit"
    ),
    row=2, col=2
)

fig.update_layout (
    title_text= "Global Superstore – Interactive Business  Dashboard",
    height=850,
    showlegend= False,
    template= "plotly_white"
)

fig.update_xaxes (tickangle=-35)
fig.show ()
```



FINAL DATA STORY: GLOBAL SUPERSTORE:

- The dataset contains 51,290 cleaned records.
- The business generated total sales of 12,642,905.00 and total profit of 1,467,457.29.
- The overall profit margin was 11.61%.
- The leading sales category was Technology.
- The leading profit category was Technology.
- The most profitable region was Central.
- Loss-making sub-categories requiring investigation: Tables

Business Priority: Protect profitable sales, investigate loss-making products, optimize discount decisions, and monitor regional performance.

10. Discuss how effective storytelling improves communication between data analysts and business stakeholders.

Data storytelling combines data analysis, visualizations, and narrative explanations to communicate meaningful insights effectively. It helps business stakeholders understand complex data through charts, graphs, and clear explanations.

In the Global Superstore dataset, visualizations help identify the highest-sales product categories, the most profitable regions, and lossmaking sub-categories. These findings allow stakeholders to understand business performance without examining thousands of individual records.

Effective data storytelling improves communication in the following ways:

1. **Simplifies complex data:** Charts and graphs make large datasets easier to understand.
2. **Highlights important insights:** Visualizations help identify trends, patterns, and business problems.
3. **Supports decision-making:** Data-driven findings help managers make informed strategic decisions.
4. **Improves stakeholder engagement:** A logical narrative explains the problem, findings, and possible solutions.
5. **Encourages actionable decisions:** Recommendations help businesses improve profitability, manage discounts, and optimize product and regional performance.

In conclusion, data storytelling transforms raw data into meaningful business knowledge. It bridges the gap between data analysts and business stakeholders by presenting evidence clearly and connecting analytical findings to practical business actions.

Questions and Answers

1. What is data storytelling? How is it different from data visualization?

Data storytelling is the process of combining data, visualizations, and narrative to communicate meaningful insights. Data visualization presents data through charts and graphs, whereas data storytelling explains the meaning of the data and suggests actions based on the findings.

2. Why is storytelling important in business analytics and decision-making?

Storytelling helps stakeholders understand complex data, identify important trends, and make informed decisions. It connects analytical findings with business objectives and supports effective problem-solving.

3. What are the essential components of an effective data story?

The essential components are data, visualization, and narrative. Data provides evidence, visualization presents patterns clearly, and narrative explains the findings and their significance. A good data story also includes actionable recommendations.

4. How do Key Performance Indicators (KPIs) enhance business reporting?

KPIs are measurable values used to evaluate business performance. They help track progress toward goals, identify performance gaps, compare results, and support data-driven decisions. Examples include total sales, profit margin, customer retention rate, and average order value.

5. Why should visualizations be arranged in a logical sequence while presenting analytical findings?

Visualizations should follow a logical sequence to make the data story clear and easy to understand. A presentation should begin with the business problem, explain the analysis, highlight the findings, and conclude with recommendations. This helps stakeholders follow the reasoning and understand the conclusions.

6. What factors should be considered while selecting visualizations for a business presentation?

The main factors are the type of data, purpose of the analysis, target audience, clarity, accuracy, and ease of interpretation. For example, bar charts compare categories, line charts show trends over time, and heatmaps display relationships or patterns.

7. Explain how dashboards and storytelling complement each other in Business Intelligence.

Dashboards display important business metrics and visualizations in one place, while storytelling explains the meaning of the displayed information. Together, they help stakeholders monitor performance, identify trends, understand business problems, and make informed decisions.

8. What challenges may arise while communicating analytical insights to non-technical stakeholders?

Common challenges include technical jargon, complex visualizations, large amounts of data, limited statistical knowledge, and difficulty connecting analytical results with business objectives. These challenges can be reduced by using simple language, clear charts, relevant examples, and practical recommendations.

9. Give two real-world examples where data storytelling has influenced business or policy decisions.

Example 1: Retail Business

A retail company analyzes sales dashboards to identify its most profitable products and underperforming regions. By presenting these findings through charts and explanations, management can improve inventory planning, optimize discounts, and focus marketing efforts on profitable products.

Example 2: Public Health

Public health authorities use visualizations of disease cases, hospital admissions, and vaccination rates to communicate health risks. These data stories help policymakers identify affected areas, allocate healthcare resources, and plan public health interventions.

10. How can effective data storytelling improve strategic planning and organizational performance?

Effective data storytelling helps organizations identify opportunities, recognize risks, set measurable goals, and allocate resources efficiently. It improves communication between teams, supports evidence-based strategies, and enables regular performance monitoring. As a result, organizations can make better decisions and improve overall performance.